1. I. Find a general solution to the given differential equation (3 points)

$$y'' - y' - 11y = 0$$

a. $y = c_1 e^{(1+3\sqrt{5})t/2} + c_2 e^{(1-3\sqrt{5})t/2}$
b. $y = c_1 e^{(2+3\sqrt{5})t/2} + c_2 e^{(2-3\sqrt{5})t/2}$
c. $y = c_1 e^{(3+3\sqrt{5})t/2} + c_2 e^{(3-3\sqrt{5})t/2}$
d. $y = c_1 e^{(3+2\sqrt{5})t/2} + c_2 e^{(3-2\sqrt{5})t/2}$
e. $y = c_1 e^{(3+2\sqrt{7})t/2} + c_2 e^{(3-2\sqrt{7})t/2}$

II. This is a case of _____ simple harmonic motion (circle one) (3 points)
ans a.,
$r^2 - r - 11 = 0$
$r = \frac{1 \pm \sqrt{1+44}}{2} = \frac{1}{2} \pm \frac{3\sqrt{5}}{2}$
two real roots: overdamped

2. Find a general solution to the initial value problem

$$y'' - 6y' + 9y = 0; \quad y(0) = 2, \quad y'(0) = \frac{25}{3}$$

a. $y = (\frac{5t}{3} - 2)e^{-3t}$
b. $y = (\frac{7t}{3} + 2)e^{-3t}$
c. $y = (\frac{5t}{3} + 2)e^{-3t}$
d. $y = (\frac{7t}{3} + 2)e^{3t}$
e. $y = (\frac{5t}{3} + 2)e^{3t}$
ans d.
$r^2 - 6r + 9 = 0$, $r = \frac{6 \pm \sqrt{36-36}}{2} = 3$, $y = c_1 e^{3t} + c_2 t e^{3t}$
$y(0) = c_1 = 2$, $y = 2e^{3t} + c_2 t e^{3t}$
$y' = 6e^{3t} + 3c_2 t e^{3t} + c_2 e^{3t}$
$y'(0) = \frac{25}{3} = 6 + c_2 = \frac{18}{3} + c_2$
$c_2 = \frac{25-18}{3} = \frac{7}{3}$

3. Solve $y'' - 4y' + 5y = 0$

a. $y = e^{-2x}(c_1 \cos(2x) + c_2 \sin(2x))$
b. $y = e^{-2x}(c_1 \cos x + c_2 \sin x)$
c. $y = e^{2x}(c_1 \cos x + c_2 \sin x)$
d. $y = e^{2x}(c_1 \cos(2x) + c_2 \sin(2x))$
e. $y = e^{-2x}(c_1 \cos(2x) + c_2 \sin(2x))$
ans c.
$r^2 - 4r + 5 = 0$, $r = \frac{4 \pm \sqrt{16-20}}{2} = 2 \pm i$
$y = e^{2x}(c_1 \cos x + c_2 \sin x)$

4. Solve $y'' - 2y' + y = 0, \quad y(0) = 5, \quad y'(0) = 10$

a. $y = e^{5x} + xe^{5x}$
b. $y = 5e^x - 5xe^x$
c. $y = 5e^x + 2xe^x$
d. $y = 5e^x - 2xe^x$
e. $y = 5e^x + 5xe^x$
ans. e
$r^2 - 2r + 1 = 0$
$(r-1)^2 = 0$
$y = c_1e^x + c_2xe^x$
$y(0) = 5 = c_1$
$y = 5e^x + c_2xe^x$
$y' = 5e^x + c_2e^x + c_2xe^x$
$y'(0) = 5 + c_2 = 10$
$c_2 = 5$

5. Can the method of undetermined coefficients be applied to the following equations?

I. $5y'' - 3y' + 2y = t^{-3}$
II. $x'' + 5x' - 3x = 3^t$
III. $y'' + 3y' - y = \sec t$
IV. $8z' - 2z = 3x^{100}e^x \cos(25x)$

a. I. yes, II. no III. yes, IV. yes
b. I. no, II. yes, III. no, IV. yes
c. I. yes, II. yes, III. no, IV. yes
d. I. yes, II. no, III. no, IV. no
e. I. no, II. no, III. no, IV. yes
ans. b
can rewrite 3^t as $e^{\ln 3t} = e^{rt}$ where $r = \ln 3$

6. Find a particular solution to the differential equation.

$$2y' + y = 3t^2$$

a. $y_p(t) = -3t - 12$
b. $y_p(t) = 3t^2 - 24t$
c. $y_p(t) = -3e^t - 12t$
d. $y_p(t) = -3te^t$
e. $y_p(t) = 3t^2 - 12t + 24$
ans e.
$2r + 1 = 0$

$r = -\frac{1}{2}$
$y_p = A_2t^2 + A_1t + A_0$
$y_p' = 2A_2t + A_1$
$4A_2t + 2A_1 + A_2t^2 + A_1t + A_0 = 3t^2$
$A_2 = 3$
$4A_2 + A_1 = 0 \Rightarrow A_1 = -12$
$2A_1 + A_0 = 0 \Rightarrow A_0 = 24$

7. Use reduction of order to find a second solution $y_2(x)$ to
$x^2y'' + 2xy' - 6y = 0; \quad y_1 = x^2$

a. $y_2 = x^3 \ln|x|$
b. $y_2 = x^3 e^x$
c. $y_2 = x^{-3}$
d. $y_2 = x^{-3} \ln|x|$
e. $y_2 = x^{-3} e^x$
$y'' + \frac{2}{x}y' - \frac{6}{x^2}y = 0;$
$y_2(x) = y_1(x) \int \frac{e^{-\int p(x)dx}}{y_1^2(x)} dx$
$y_2(x) = x^2 \int \frac{e^{-\int \frac{2}{x}dx}}{x^4} dx$
$y_2(x) = x^2 \int \frac{e^{-2\ln x}}{x^4} dx$
$y_2(x) = x^2 \int \frac{x^{-2}}{x^4} dx$
$y_2(x) = x^2 \int x^{-6} dx$
$y_2(x) = x^2 \cdot \frac{x^{-5}}{-5} = -\frac{x^{-3}}{5}$
ans. c.

8. Solve the Cauchy-Euler equation $x^2\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} - 4y = 0$
a. $y = c_1e^{-x} + c_2e^{4x}$
b. $y = c_1x^{-1} + c_2x^4$
c. $y = c_1x + c_2x^{-4}$
d. $y = c_1e^x + c_2e^{-4x}$
e. $y = c_1xe^x + c_2xe^{-4x}$
(ans b)
$r^2 + (-2-1)r - 4 = 0$
$r^2 + -3r - 4 = 0$
$(r-4)(r+1) = 0$
$r = -1, 4$
$y = c_1x^{-1} + c_2x^4$

9. Find the form of a particular solution for $y'' - 2y' + y = 7e^t \cos t$. Do not solve for the coefficients.
a. $y_p(t) = (A_0 \cos t + B_0 \sin t)te^t$
b. $y_p(t) = (A_0 \cos t + B_0 \sin t)t^2e^t$
c. $y_p(t) = (A_0 \cos t + B_0 \sin t)e^t$
d. $y_p(t) = [(A_1t + A_0)\cos t + (B_1t + B_0)\sin t]e^t$
e. $y_p(t) = [(A_1t + A_0)\cos t + (B_1t + B_0)\sin t]te^t$
(ans c)
$r^2 - 2r + 1 = 0$
$r = \frac{2 \pm \sqrt{4-4}}{2} = 1$
$1 + i$ is not a root of the auxiliary equation.
$y_p(t) = (A_0 \cos t + B_0 \sin t)e^t$

1. Solve $y'' - 2y' + y = \frac{e^x}{1+x^2}$ using the method of variation of parameters and fill in the blanks (each: 1 pt answer, 1 pt supporting work).

The auxiliary equation is $m^2 - 2m + 1 = (m-1)^2 = 0$, so $y_c = c_1 e^x + c_2 x e^x$ and

$$W = \begin{vmatrix} e^x & xe^x \\ e^x & xe^x + e^x \end{vmatrix} = e^{2x}.$$

Identifying $f(x) = e^x / \left(1 + x^2\right)$ we obtain

$$u_1' = -\frac{xe^x e^x}{e^{2x}\left(1+x^2\right)} = -\frac{x}{1+x^2}$$

$$u_2' = \frac{e^x e^x}{e^{2x}\left(1+x^2\right)} = \frac{1}{1+x^2}.$$

$W_1 = -\frac{xe^{2x}}{1+x^2}$
$W_2 = \frac{e^{2x}}{1+x^2}$

Then $u_1 = -\frac{1}{2}\ln\left(1+x^2\right)$, $u_2 = \tan^{-1} x$, and

$$y = c_1 e^x + c_2 x e^x - \frac{1}{2} e^x \ln\left(1+x^2\right) + x e^x \tan^{-1} x.$$

2. Solve $y'' + 2y' = 2x + 5 - e^{-2x}$ using the method of undetermined coefficients. Call your constants A, B, C, etc. Fill in the blanks (each: 1 pt answer, 1 pt supporting work).

-2 is a root of the char. eqn. so s=1.

From $m^2 + 2m = 0$ we find $m_1 = -2$ and $m_2 = 0$. Then $y_c = c_1 e^{-2x} + c_2$ and we assume $y_p = Ax^2 + Bx + Cxe^{-2x}$. Substituting into the differential equation we obtain $2A + 2B = 5$, $4A = 2$, and $-2C = -1$. Then $A = \frac{1}{2}$, $B = 2$, $C = \frac{1}{2}$, $y_p = \frac{1}{2}x^2 + 2x + \frac{1}{2}xe^{-2x}$, and

$$y = c_1 e^{-2x} + c_2 + \frac{1}{2}x^2 + 2x + \frac{1}{2}xe^{-2x}.$$

roots of char. eqn. = _____

$y_c =$ _____

form of particular soln $y_p =$ _____

$A =$ _____

$B =$ _____

$C =$ _____

particular soln $y_p =$ _____

$y_g =$ _____

3. Solve $4y'' + 4y' + 17y = 0, \quad y(0) = -1, \quad y'(0) = 2$. Call your constants c_1 and c_2. Fill in the blanks (each: 1 pt answer, 1 pt supporting work).

roots of the auxiliary equation $4m^2 + 4m + 17 = 0$ are $m_1 = -\frac{1}{2} + 2i$ and $m_2 = -\frac{1}{2} - 2i$. $y = e^{-x/2}(c_1 \cos 2x + c_2 \sin 2x)$. Applying the condition $y(0) = -1$, we see from $e^0(c_1 \cos 0 + c_2 \sin 0) = -1$ that $c_1 = -1$. Differentiating $y = e^{-x/2}(-\cos 2x + c_2 \sin 2x)$ and then using $y'(0) = 2$ gives $2c_2 + \frac{1}{2} = 2$ or $c_2 = \frac{3}{4}$. solution of the IVP is $y = e^{-x/2}(-\cos 2x + \frac{3}{4} \sin 2x)$.

solution is oscillatory, but $y \to 0$ as $x \to \infty$ and $|y| \to \infty$ as $x \to -\infty$.

should be amplitude, not y, going to infinity.

roots of char. eqn. = _____

$y_c =$ _____

$c_1 =$ _____

$c_2 =$ _____

$y =$ _____

$\lim_{x\to\infty} y(x) =$ _____

As $x \to -\infty$, $|y| \to$ _____

should be amplitude, not y, going to infinity

Form B multiple choice

1. c.
2. b.
3. c.
4. e.
5. b.
6. e.
7. a., overdamped
8. d.
9. c.